

Education

- 2023.09 – Present **Ph.D. & M.S. in Computational Mathematics**, *Stanford University*.
Thesis: The exchange-correlation functional: exact definition, computational access, and structured approximation.
Advisor: Prof. Lexing Ying.
- 2019.10 – 2023.08 **Ph.D. & M.S. in Chemical Physics**, *University of Chicago*.
Thesis: Multiscale methods for quantum many-body systems.
Advisor: Prof. Yuehaw Khoo.
- 2015.09 – 2019.07 **B.S. in Physics**, *University of Chinese Academy of Sciences*.
Thesis: Density matrix renormalization group applied to quantum chemical calculations.
Advisor: Prof. Tao Xiang.
- 2015.09 – 2019.07 **B.S. in Chemistry**, *University of Chinese Academy of Sciences*.
Thesis: GPU acceleration of matrix project state based hierarchical equations of motion.
Advisor: Prof. Qiang Shi.

Work Experience

- 2025.06 – Present **Research Intern**, *ByteDance Ltd.*.
Project: Robust machine-learning quantum chemistry methods for molecular science.
Advisor: Dr. Yixiao Chen, and Dr. Weiluo Ren.
- 2021.06 – 2021.08 **Research Intern**, *Flatiron Institute*.
Project: Low-rank Green's function representations applied to dynamical mean-field theory.
Advisor: Dr. Jason Kaye, Dr. Kun Chen, and Dr. Olivier Parcollet.

Preprints

- 2026 **Nan Sheng**. A unified variational framework for the inverse kohn-sham problem. *arXiv preprint arXiv:2603.23452*, 2026.
- 2026 **Nan Sheng**. Exact density-functional theory as parallel ensemble variational hierarchies: from lieb's formulation to kohn-sham theory. *arXiv preprint arXiv:2603.23399*, 2026.
- 2025 Titouan Duston, Shuo Xin, Yang Sun, Daoguang Zan, Aoyan Li, Shulin Xin, Kai Shen, Yixiao Chen, Qiming Sun, **Nan Sheng**, Ge Zhang, et al. Ainsteinbench: Benchmarking coding agents on scientific repositories. *arXiv preprint arXiv:2512.21373*, 2025.
- 2023 **Nan Sheng**, Siyao Yang, and Yuehaw Khoo. Multiscale initialization methods for high-dimensional eigenvalue problems. *In preparation*, October 2023.

Publications

- 2025 Xun Tang, **Nan Sheng**, and Lexing Ying. Solving high-dimensional Hamilton-Jacobi-Bellman equations with functional hierarchical tensor. *arXiv preprint arXiv:2408.04209*, To appear in *SIAM Journal on Scientific Computing*, June 2025.

- 2023 **Nan Sheng**, Alexander Hampel, Sophie Beck, Olivier Parcollet, Nils Wentzell, Jason Kaye, and Kun Chen. Low-rank Green's function representations applied to dynamical mean-field theory. *Phys. Rev. B*, volume 107, page 245123. American Physical Society, June 2023.
- 2023 Benchen Huang, **Nan Sheng**, Marco Govoni, and Giulia Galli. Quantum simulations of Fermionic Hamiltonians with efficient encoding and ansatz schemes. *J. Chem. Theory. Comput.*, volume 19, pages 1487–1498. American Chemical Society, February 2023.
- 2022 Christian Vorwerk, **Nan Sheng**, Marco Govoni, Benchen Huang, and Giulia Galli. Quantum embedding theories to simulate condensed systems on quantum computers. *Nature Comput. Sci.*, volume 2, pages 424–432. Nature Publishing Group, July 2022.
- 2022 **Nan Sheng**, Christian Vorwerk, Marco Govoni, and Giulia Galli. Green's function formulation of quantum defect embedding theory. *J. Chem. Theory. Comput.*, volume 18, pages 3512–3522. American Chemical Society, June 2022.
- 2021 He Ma, **Nan Sheng**, Marco Govoni, and Giulia Galli. Quantum embedding theory for strongly correlated states in materials. *J. Chem. Theory. Comput.*, volume 17, pages 2116–2125. American Chemical Society, April 2021.
- 2020 He Ma, **Nan Sheng**, Marco Govoni, and Giulia Galli. First-principles studies of strongly correlated states in defect spin qubits in diamond. *Phys. Chem. Chem. Phys.*, volume 22, pages 25522–25527. The Royal Society of Chemistry, November 2020.

Talks

- 2025 Field-Theoretic and Tensor Network Methods for Scientific Computing, Seed Research Intern Seminar, *Bytedance Ltd.*.
- 2024 Introduction to computational quantum physics, Workshop on Applied Math for Quantum Physics, *The University of Chicago*.
- 2023 Introduction to computational quantum physics, Student Applied Math Seminar, *The Ohio State University*.
- 2023 Green's function formulation of quantum defect embedding theory, *APS March Meeting*.
- 2023 Quantum simulations of Fermionic Hamiltonians with efficient encoding and ansatz schemes, *APS March Meeting*.
- 2022 Extrinsic and intrinsic defects in MgO and CaO as potential spin-qubit candidates, *APS March Meeting*.
- 2022 An exact double counting scheme for quantum defect embedding theory, *APS March Meeting*.
- 2021 Accelerating dynamical mean-field calculations using the discrete Lehmann representation, CCQ Summer Intern Seminar, *Flatiron Institute*.
- 2021 First-principles studies of strongly correlated states in defect spin qubits in diamond, *APS March Meeting*.
- 2021 Coupling interoperable software for quantum simulations of materials, *APS March Meeting*.

Refereeing Activities

Reviewer: Physical Review Letters, Physical Review X, PRX Quantum, Physical Review B, Physical Review A, Physical Review Research, Chinese Physics B, Computer Physics Communications, Physics of Fluids, Machine Learning: Science and Technology, Quantum Reports, Physical Chemistry Chemical Physics

Teaching Activities

- 2026 **Vector Calculus for Engineers**, *Stanford University*, course assistant.
- 2025, 2026 **Calculus**, *Stanford University*, course assistant.

- 2025 **Computational Methods of Applied Mathematics**, *Stanford University*, course assistant.
2020 **Comprehensive General Chemistry**, *University of Chicago*, teaching assistant.
2019, 2020 **Organic Chemistry**, *University of Chicago*, teaching assistant.

Fellowships & Awards

- 2023 **ICME Fellowship**, *Stanford University*.
2019 **McCormick Fellowship**, *University of Chicago*.
2019 **Excellent Graduate of Beijing**, *Chinese Ministry of Education*, 2 out of 35.
2019 **Excellent Graduate**, *University of Chinese Academy of Sciences*, 3 out of 35.
2018 **Study Abroad Scholarship**, *University of Chinese Academy of Sciences*, 2 out of 35.
2018 **Tang Lixin Scholarship**, *University of Chinese Academy of Sciences*, 1 out of 35.
2016, 2017, 2018 **National Encouragement Scholarship**, *Chinese Ministry of Education*, 2 out of 35.
2016, 2017, 2018 **Academic Excellence Scholarship**, *University of Chinese Academy of Sciences*.

Technical Skills

Research: Computational science, Numerical methods, Mathematical modeling, Machine learning
Programming: Python, MATLAB, Fortran, C/C++, Bash, L^AT_EX, MPI, CUDA
Software: CVXPY, JAX, PySCF, Qiskit, Quantum Espresso, Gaussian, ORCA, TRIQS, Wannier90